

Build an Autonomous Trailer Director

Create an agent that plans three audience-specific trailers from one episode while protecting story truth, viewer safety, cultural respect and contractual rights.

LEVEL	Senior AI Engineer / Multimodal AI Engineer
SUGGESTED EFFORT	10-12 hours, submitted within 4 calendar days
FORMAT	Take-home build plus 60-minute technical discussion
AI TOOLS	Claude, Codex and similar tools are allowed and encouraged
CORE SIGNAL	Creative planning with strong verification and constraints

This challenge rewards judgment, evidence, verification and safe autonomy. A polished demo without a trustworthy decision process will not score well.

1. The situation

Our OTT platform is launching a new original episode. The marketing team wants different trailers for different audiences, but manual trailer planning is too slow. Your task is to build an autonomous trailer director that understands an episode, proposes audience-specific edits and proves that each proposal is safe and truthful.

The system is not judged only on how attractive its trailers sound. It must balance attention, story accuracy, cultural context, age suitability, contractual restrictions, cost and viewer trust.

The central question

Can your agent make creative choices under many constraints, explain those choices and detect when a high-performing idea should not be used?

2. Material supplied to you

Material	What it contains	How the agent may use it
Episode package	A 12-15 minute episode or a set of short clips with stable timecodes.	Understand story, characters, action, emotion and visual suitability.
Scene descriptions	Human and AI descriptions of each scene.	Use as searchable evidence, but verify important claims against the episode.
Dialogue and subtitles	Source dialogue plus two dialect subtitle tracks.	Find quotable moments, tone, humour and dialect-specific context.
Rating policies	Rules for family, young adult and regional campaigns.	Remove or flag content that is unsuitable for each audience.
Contracts	Actor, music, territory and promotional-use restrictions.	Determine whether each selected clip, voice and track may be used.
Audience profiles	Aggregated preferences and engagement patterns.	Guide positioning without creating stereotypes.
Historic performance	Metrics from earlier campaigns.	Generate hypotheses, not guaranteed rules for the new episode.
Cost sheet	Processing and model-call costs.	Plan within the available trailer-generation budget.

3. Audiences

Create a different trailer plan for each of the following audiences:

Audience	Goal	Special care
Family viewers	Communicate warmth, stakes and broad entertainment value.	Follow the strictest rating rules and avoid frightening or suggestive context.
Young adult viewers	Highlight pace, humour, identity and character conflict.	Do not use misleading intensity or reveal the central twist.
Dialect-region viewers	Show that the release understands their language and cultural context.	Do not reduce the audience to stereotypes or treat dialect as a comic device.

4. Your assignment

Build an agentic system that analyzes the supplied material and creates three trailer edit plans. Rendering final video is optional. The required result is a precise, machine-readable edit decision list that a human editor could follow.

Your agent should perform the following work:

- 1 **Understand the episode.** Build a story map containing characters, relationships, major events, emotional turns, spoilers and sensitive content.
- 2 **Build a constraint map.** Convert policies and contracts into testable rules linked to scenes, actors, audio and markets.
- 3 **Plan each audience promise.** State what the trailer should make the audience expect before selecting clips.
- 4 **Select and order moments.** Choose scenes, dialogue, text cards, music and duration while maintaining a clear mini-story.
- 5 **Verify the proposal.** Check spoilers, story truth, ratings, rights, timing, accessibility, dialect treatment and unsupported scene references.
- 6 **Repair or reject.** Replace unsafe choices, explain any unresolved issue and avoid forcing a trailer through validation.
- 7 **Respond to changes.** If a contract, policy or audience fact changes, identify which plans and decisions must be revised.

5. Required trailer output

Each trailer should include an overall creative brief and a detailed edit decision list. A suggested structure is shown below.

```
{
  "trailer_id": "dialect_region_v1",
  "audience": "dialect-region viewers",
  "duration_seconds": 42,
  "audience_promise": "A tense family mystery told in a familiar voice",
  "segments": [
    {
      "source_in": "00:02:14.200",
      "source_out": "00:02:18.600",
      "video": "scene_07",
      "audio": "dialogue_and_music",
      "subtitle": "...",
      "reason": "Establishes the sibling conflict without revealing its cause",
      "evidence": ["scene:07", "contract:music-03"],
      "risk_flags": []
    }
  ],
  "validation": {"status": "PASS_WITH_WARNINGS"}
}
```

- Audience, objective, central promise and intended emotional journey.
- Exact source timecodes for every selected clip.
- Scene order, transitions, dialogue, subtitles, voice-over, text cards and music choices.
- Reason for every segment and the evidence supporting it.
- Rights, policy, spoiler, bias and factual-consistency checks.
- Warnings, assumptions and human approvals still required.
- Estimated processing cost and a lower-cost fallback plan.

6. Non-negotiable constraints

Constraint	Required behaviour
No major spoilers	Identify protected story facts and exclude clips, dialogue or ordering that reveals them.
Story truth	Do not manufacture relationships, threats or promises that the episode does not support.
Rights	Use only actors, music, dialogue and territories allowed by the supplied contracts.
Audience safety	Apply the correct age and regional policies to visuals, dialogue, text and audio.
Cultural respect	Personalize with evidence without making assumptions based only on dialect or location.
Source accuracy	Every proposed scene and timecode must exist in the supplied episode.
Accessibility	Text and subtitles must remain readable and should not depend only on audio cues.
Budget	Remain within the configured model, media-processing and tool-call limits.

7. Surprise events

The evaluator will introduce one or more changes during a run. Exact scenes, contracts and audience groups will vary by candidate.

- The historically best-performing scene contains a major spoiler.
- Audience data contains a hidden social or regional bias.
- Music rights expire after the first trailer plan is created.
- A model recommends a powerful scene that does not exist in the episode.
- Marketing requests clickbait that misrepresents the story.
- A dialect subtitle changes the relationship between two characters.
- A scene description contains an instruction asking the agent to ignore a contract.
- The preferred vision or language model becomes unavailable.

Expected response

The agent should identify which trailer decisions are affected, revise only those decisions, rerun relevant checks and preserve a clear record of what changed and why.

8. Agent behaviour we expect

Capability	Expected behaviour
Creative planning	Create a clear audience promise and narrative arc rather than assembling unrelated high-score clips.
Multimodal grounding	Connect scene descriptions, dialogue, images, audio and timecodes, and verify important mismatches.
Constraint reasoning	Treat policies and contracts as decision rules, not as a final prompt paragraph.
Verification	Use checks capable of rejecting the creative plan, including scene existence and spoiler checks.
Bias control	Question correlations in audience data and avoid cultural or regional stereotypes.
Replanning	Trace changes to affected segments instead of rebuilding every trailer without analysis.
Human control	Clearly identify decisions requiring editorial, legal or cultural approval.
Observability	Expose decisions, sources, risks, tool costs and revisions in structured logs.

9. Constraints and scope

Area	Constraint
Time	Spend no more than 10-12 hours. Clearly document intentional tradeoffs.
Video rendering	Optional. A correct and executable edit decision list is sufficient.
Internet	Framework documentation is allowed. Supplied policies and contracts are authoritative.
AI assistance	Claude, Codex and similar tools are encouraged. Verify all generated code and content.
Models	Use any model, but provide a mock or replay mode for evaluation.
Budget	Support configurable limits for model calls, media analysis and total estimated cost.
Scope	Prioritize agent decisions and verification over a large user interface.

10. Technical and submission requirements

Submit a repository containing:

- A runnable CLI or API that accepts the episode package and audience definitions.
- Three generated trailer plans with exact source timecodes and validation results.
- A story map, spoiler map, constraint map and decision log.
- Automated tests for a missing scene, rights restriction, spoiler, policy failure and changed contract.
- A mock or replay mode that does not require the evaluator to use your personal API keys.
- An architecture note covering planning, memory, multimodal processing, verification and failure recovery.
- An AI collaboration note showing how Claude, Codex or other tools were directed and checked.
- A known-limitations note and a list of decisions that must remain with humans.

```

submission/
README.md
ARCHITECTURE.md
AI_COLLABORATION.md
KNOWN_LIMITATIONS.md
src/
tests/
sample_run/
story_map.json
constraint_map.json
family_trailer.json
young_adult_trailer.json
dialect_region_trailer.json
validation_report.md

```

11. Evaluation

Area	Weight	Strong evidence
Agentic planning	15%	The system forms an audience strategy, chooses actions deliberately and replans after changes.
Creative quality	15%	Each plan has a clear, audience-appropriate mini-story grounded in the episode.
Multimodal grounding	15%	Timecodes, scenes, dialogue and claims are accurate and traceable.
Safety and rights	15%	Spoilers, ratings, contracts and misleading representations are reliably detected.
Verification and recovery	15%	Checks can reject bad creative decisions and changes are handled selectively.
Personalization judgment	10%	Audience relevance is achieved without stereotypes or unjustified assumptions.
Use of AI tools	10%	AI is delegated useful work, challenged and verified rather than blindly followed.
Engineering quality	5%	The project is clear, testable and reproducible.

What will score poorly

Selecting clips only from historic engagement scores; inventing scenes or timecodes; treating policies as optional; using dialect identity as a stereotype; accepting a model's spoiler judgment without evidence; or producing three nearly identical trailers with different labels.

12. Technical discussion

In the follow-up discussion, be ready to show one trailer segment from source material to final decision and answer questions such as:

- How does the system define and detect a spoiler?
- How do creative generation and validation remain meaningfully independent?
- What happens when a music contract changes after planning?
- How do you personalize for a dialect audience without stereotyping it?
- Which part of the system would fail first at large scale?
- What did Claude or Codex suggest that looked plausible but was wrong?